

Name:

**Instructions:**

- All answers must be written clearly.
- Calculators are not allowed. Use factorial and permutation or combination notation as necessary. **Don't use decimals.**
- Be sure to erase or cross out any work that you do not want graded.
- Circle your answer to distinguish it from intermediate steps where applicable.
- If you need extra space, you may use the back sides of the exam pages (if you do, please write me a note so that I know where to look).
- With the exception of Question 1, you must explain your reasoning to receive full credit.

[illegible]

1. Answer **TRUE** if the statement is always true, otherwise answer **FALSE**. No justification is required. (18)

a) The proposition  $(p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$  is a contingency.

b) The proposition  $\exists x P(x) \vee \forall y Q(y)$  is equivalent to the proposition  $\exists x (P(x) \vee Q(x))$ .

c) If  $\forall x P(x) \vee \forall y Q(y)$ , then  $\forall x (P(x) \vee Q(x))$ .

d) The cardinality of  $\mathcal{P}(\{\emptyset\})$  is 1.

e) If  $A \subseteq B$ , then  $A \times B \subseteq B \times B$ .

f) The function  $f(x) = |x| + 1$  with domain  $\mathbb{R}$  and codomain  $[0, \infty)$  is one-to-one.

g) If  $A$  and  $B$  are disjoint events, then  $P(A \cup B) = P(A) + P(B)$ .

h) If  $E$  and  $F$  are disjoint events, then  $E$  and  $F$  are independent.

i) If  $A$  and  $B$  are events with  $P(B) > 0$ , then  $P(\overline{A}|B) = 1 - P(A|B)$ .

2. Consider the propositions:

(6)

- $p$  : You have the flu
- $q$  : You miss the final exam
- $r$  : You pass the course.

Express the following compound propositions as English sentences:

a)  $p \rightarrow q$

b)  $\neg q \leftrightarrow r$

c)  $(p \wedge q) \vee (\neg q \wedge r)$

3. Construct a truth table for the proposition  $(p \rightarrow q) \leftrightarrow (\neg p \vee q)$ . Is this proposition a tautology, contradiction, or contingency?

(5)

4. Consider the following predicates where the domain is all SLU students: (7)

- $K(x, y) : x$  knows  $y$ ,
- $M(x) : x$  is a math major,
- $B(x) : x$  is on the basketball team.

Suppose that Bob is a SLU student. Use quantifiers along with these predicates to express the following propositions:

- Every math major knows everyone on the basketball team.
- Someone on the basketball team knows everyone except for Bob.
- Write the negation of your answer to part a) so that every negation sign is to the immediate left of a predicate.
- Express your answer to part c) as an English sentence.

5. Use the rules of inference to show that the four premises  $r \rightarrow s$ ,  $\neg p$ ,  $q \rightarrow r$ , and  $p \vee q$  lead to the conclusion  $s$ . (5)

6. Suppose  $A$  and  $B$  are subsets of the universal set  $U$ . (4)

a) Illustrate  $A \cap \overline{B}$  with a Venn diagram.

b) Illustrate  $A - \overline{B}$  with a Venn diagram.

7. Consider functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  and  $g : \mathbb{R} \rightarrow \mathbb{R}$ . (5)

a) Give a **precise** definition of what it means for  $f(x)$  to be  $\Omega(g(x))$ .

b) Verify that  $f(x) = x^2 - 1$  is  $\Omega(x)$  by showing that it satisfies the definition from part a).

8. a) What is the definition of an even integer? (6)
- b) What is the definition of an odd integer?
- c) Suppose that  $m$  and  $n$  are integers. Prove that if  $mn$  is even, then  $m$  or  $n$  is even. Be sure to clearly indicate which method of proof that you use.
9. Consider the recurrence relation  $a_n = a_{n-1} + 2$  with initial condition  $a_1 = -1$ . (6)
- a) List the first five terms of the sequence defined by this recurrence relation and initial condition.
- b) Conjecture a formula for  $a_n$  in terms of  $n$ . Check that your conjecture agrees with part a).
- c) Use mathematical induction to prove that your formula from part b) holds for all  $n$ .

10. A ballroom dancing class has 24 students, of which 11 are men and 13 are women. (6)
- a) How many ways can a group of 6 men and 6 women be selected?
  - b) How many ways can one of these groups from part a) be paired off into teams of one woman and one man?
  - c) How many ways can a group of 4 be selected if there must be more women than men in the group?
11. A can of soda in a vending machine costs 85 cents. Assuming that coins of equal value are identical, in how many ways can a customer put in: (6)
- a) eighty-five pennies one coin at a time.
  - b) one half dollar, one quarter, and one dime one coin at a time.
  - c) two quarters, two dimes, and three nickels one coin at a time.

12. Suppose a fair coin is flipped five times in succession. (5)

a) How many outcomes are in the sample space?

b) How many outcomes have at least 2 heads?

c) What is the probability of getting at least 2 heads?

13. Suppose a fair 6-sided die is rolled twice in succession. Let  $A$  denote the event *first roll is odd* and  $B$  denote the event *sum of numbers is equal to 7*. (6)

a) Calculate  $P(A)$  and  $P(B)$ .

b) Calculate  $P(A \cap B)$  and  $P(B|A)$ .

c) Are  $A$  and  $B$  independent? Why or why not?